

DESIGN AND DEVELOPMENT OF SENTIMENT FRAMEWORK USING NLP

Macro Project 2- Report

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DECLARATION

I hereby declare that the work entitled **“DESING AND DEVELOPMENT OF SENTIMENT FRAMEWORK USING NLP”** is my work, conducted under the supervision of **Dr.Tej Singh, Assistant professor**, during the session January-June 2026. The report submitted by me is a record of bona fide work carried out by our team.

I further declare that the work reported in this report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge and belief.

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ABSTRACT

The **Sentiment Analysis System** is a project we have made to help users and organizations understand people's opinions and feedback easily, and the admin can see all the results and analysis using this system. It helps to analyse text data like reviews, comments, feedback and classify them into positive, negative, or neutral sentiments as what their meaning is.

Before some years, most companies and platforms manually read, review and feedback to understand user opinions, which always took a lot of time and due to this chances of mistakes also increases. This project makes the analysis faster and easier.

The system allows the user to enter any text and get the sentiment result instantly. It also helps to analyse large amounts of data and understand overall trends and opinions of users.

This project is made using Streamlit for the frontend and Python with machine learning and NLP techniques like TF-IDF and classification algorithms Logistic Regression and to automate tasks like processing text and predicting sentiment. It is simple, user-friendly, and helps in making better decisions based on user feedback.

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CHAPTER 1: INTRODUCTION

The Sentiment Analysis System is a project we made to help users and organizations to manage and understand people's opinions easily according to their needs.

In many old methods, feedback and reviews are read manually or stored in files without proper analysis. This always takes a lot of time and can make mistakes. To solve this problem, this project helps to analyse all the text data in a very easy way and gives results quickly.

This system is made using Python with machine learning and NLP techniques and Streamlit. With the help of this, many tasks like processing user input, analysing text, and predicting sentiment can be done automatically with just one click.

The main aim of this project is to:

1. Save time and reduce manual work of reading and analysing text data.
2. Keep all feedback analysis at one place.
3. Avoid mistakes and confusion in understanding user opinions.
4. Help users and organizations to make better decisions more efficiently.

CHAPTER 2: LITERATURE SURVEY

In the earlier days, most companies and organizations used manual methods to understand customer opinions by reading reviews and feedback one by one. This process was slow, time-taking, and sometimes caused mistakes to understand the actual sentiment. Later, many started using Excel sheets to store feedback data, which made work slightly faster but still required manual analysis which took a lot of effort.

Some developers then created computer-based programs to analyse text data, which helped in processing feedback more easily, but these were limited and not very good. As technology got improved, machine learning models were introduced that allowed automatic sentiment analysis with good accuracy, but they required technical knowledge and proper setup.

Nowadays, many platforms use advanced AI models and real-time systems to analyse user opinions from social media, reviews, and comments quickly. From studying all these methods pros and cons, it is clear that while there are many ways to perform sentiment analysis, many users still need a simple and user-friendly system.

Our project aims to provide such a system using machine learning and Streamlit that makes sentiment analysis faster, more accurate, and easy to use.

CHAPTER 3: METHODOLOGY

1. Problem Identification

- Organizations face problems in manually understanding large amounts of text data like reviews, comments, and feedback
 - It takes time and may cause errors in the actual sentiment of users.
-

2. Data Collection

- Text data collected from available datasets reviews or feedback data.
- Ideas were taken from existing sentiment models and resources.

3. System Design

- The system was designed to include main modules like Text Input, Preprocessing, Feature Extraction, Model Prediction, and Result Display.
- The workflow is plan to be simple and user friendly so that any user can useit.

4. Implementation

- The project was developed using Python and Streamlit for the frontend interface.
- Machine learning and NLP techniques like TF-IDF, Logistic Regression
- Input system and prediction output were created for easy use.

5. Testing

- The system was tested by giving different types of text inputs to check if predictions were correct.
- Errors were fixed to make the system more accurate and gives correct result.

6. Result and Analysis

- The final system successfully analysed text data and predicted sentiment in less time.
- It reduced manual effort and made sentiment analysis faster and more accurate.

CHAPTER 4: IMPLEMENTATION

In this part of the project, we have successfully developed both the front-end and back-end of the Sentiment Analysis System using Python, Streamlit, and machine learning techniques. This system provides a complete solution for analysing text data and predicting sentiment

Core Components

1. Streamlit Interface

- Using Streamlit we have designed a simple and easy to use interface for users.
 - Created input box for entering text and button for predicting sentiment.
 - Displayed results as Positive, Negative or Neutral.
-

2. Data Preprocessing

- Implemented text cleaning techniques like lower casing, removing punctuation, numbers
 - Used tokenization to split text into words
 - Ensured that input text is properly processed before prediction.
-

3. Feature Engineering (TF-IDF)

- Used TF-IDF Vectorization to convert text into numerical value.
 - Important words were given more importance compared to common words.
 - Stored the vectorizer for re use during prediction.
-

4. Machine Learning Models

- Implemented Logistic Regression for classification.
- Trained models on dataset.
- Selected the best performing model for final prediction.

5. Prediction System

- Added functionality to take user input and.
- Used trained model along with preprocessing .
- Displayed output results in real-time.

6. Model Integration and Storage

- Saved trained model using pickle for reuse.
- Loaded the model in Streamlit app for fast predictions.
- Ensured smooth integration between preprocessing, model, and UI.

7. System Reliability

- Tested system with different types of inputs to check accuracy.
- Ensured consistent results and proper error handling.
- Made the system simple, fast, and user-friendly.

CHAPTER 5: RESULT



Sentiment Analyzer

Analyze the sentiment of any text - Positive, Negative, or Neutral

Enter your text below:

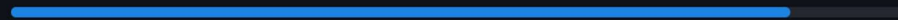
The movie was good

Analyze Sentiment



Sentiment: **POSITIVE**

Confidence Score: 87.5%



> Why did we predict this?

CHAPTER 6: CONCLUSION

The **Sentiment Analysis System** was created to make understanding user opinions faster and easier. Instead of reading reviews or feedback manually, this system allows all text data to be analysed directly through a simple interface.

Before this, most people analysed feedback manually, which took extra time and often caused mistakes in understanding the actual sentiment. With this system, everything—from entering text to getting sentiment results—can be done in just a few clicks.

The project gives users an easy way to input any text and instantly find whether the sentiment is positive, negative, or neutral. It also helps in analysing large amounts of feedback efficiently, saving time and effort.

The platform is simple to use and doesn't require much technical skill. It processes texts automatically using machine learning technique and provide accurate results. Overall, this system helps in better understanding the opinions, improves decision-making, and makes analysing feedback much more easy.

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